Module 2.3 Rearranging formulas

Printable flashcards

38 cards: equations, key ideas, quick-check questions and problems.
Cut, fold and review; or use the scheduled version in the app.

How to use

Printable flashcards

Each row is one card: the prompt on the left, the answer on the right. Print single-sided, cut along the rows, and fold each strip down the middle so the answer sits on the back. The same cards are in the app, where they are scheduled for review automatically.

Module 2.3P · Rearranging formulas (pilot)

REARRANGE Make a the subject (Kinematics: final velocity): $v = u + at$	$a = \frac{v-u}{t}$
REARRANGE Make x the subject (Straight line): $y = mx + c$	$x = \frac{y-c}{m}$
REARRANGE Make w the subject (Perimeter of a rectangle): $P = 2l + 2w$	$w = \frac{P-2l}{2}$
REARRANGE Make C the subject (Temperature conversion): $F = \frac{9}{5}C + 32$	$C = \frac{5(F-32)}{9}$
REARRANGE Make u the subject (Distance under uniform acceleration): $s = \frac{(u+v)t}{2}$	$u = \frac{2s}{t} - v$
REARRANGE Make T the subject (Ideal gas law): $PV = nRT$	$T = \frac{PV}{nR}$
REARRANGE Make v the subject (Kinetic energy): $E = \frac{1}{2}mv^2$	$v = \sqrt{\frac{2E}{m}}$

REARRANGE Make g the subject (Simple pendulum): $T = 2\pi\sqrt{\frac{L}{g}}$	$g = \frac{4\pi^2 L}{T^2}$
REARRANGE Make r the subject (Newtonian gravity): $F = \frac{Gm_1m_2}{r^2}$	$r = \sqrt{\frac{Gm_1m_2}{F}}$
REARRANGE Make r the subject (Area of a circle): $A = \pi r^2$	$r = \sqrt{\frac{A}{\pi}}$
REARRANGE Make v the subject (Thin lens equation): $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$	$v = \frac{uf}{u-f}$
REARRANGE Make r the subject (Volume of a sphere): $V = \frac{4}{3}\pi r^3$	$r = \sqrt[3]{\frac{3V}{4\pi}}$
REARRANGE Make M the subject (Kepler's third law (Newton's form)): $T^2 = \frac{4\pi^2}{GM}a^3$	$M = \frac{4\pi^2 a^3}{GT^2}$
REARRANGE Make x the subject (Unknown on both sides): $ax + b = cx + d$	$x = \frac{d-b}{a-c}$
REARRANGE Make x the subject (Inverting a rational function): $y = \frac{x+2}{x-3}$	$x = \frac{3y+2}{y-1}$
REARRANGE Make R2 the subject (Two resistors in parallel): $R = \frac{R_1R_2}{R_1+R_2}$	$R_2 = \frac{RR_1}{R_1-R}$

REARRANGE Make vs the subject (Doppler effect, moving source): $f' = \frac{fv}{v - v_s}$	$v_s = \frac{v(f' - f)}{f'}$
REARRANGE Make x the subject (Bracketed unknown): $a(x - b) = cx$	$x = \frac{ab}{a - c}$
REARRANGE Make t the subject (Radioactive decay): $N = N_0 e^{-kt}$	$t = \frac{\ln(N_0/N)}{k}$
REARRANGE Make H the subject (Acidity): $\mathrm{pH} = -\log_{10}[\mathrm{H}^+]$	$[\mathrm{H}^+] = 10^{-\mathrm{pH}}$
REARRANGE Make t the subject (Compound growth): $A = P(1 + r)^t$	$t = \frac{\ln(A/P)}{\ln(1 + r)}$
REARRANGE Make x the subject (General exponential): $y = ab^x$	$x = \frac{\ln(y/a)}{\ln b}$
REARRANGE Make I the subject (Sound level in decibels): $L = 10\log_{10}\left(\frac{I}{I_0}\right)$	$I = I_0 \cdot 10^{L/10}$
REARRANGE Make t the subject (Newton's law of cooling): $T = T_s + (T_0 - T_s)e^{-kt}$	$t = \frac{1}{k}\ln\left(\frac{T_0 - T_s}{T - T_s}\right)$
SPOT THE IDEA You time a pendulum at several lengths L. Which graph gives a straight line through the origin, and what is its gradient?	T^2 against L; gradient $4\pi^2/g$. Compare with y = m x: y is T squared, x is L, and m is 4 pi squared over g.

SPOT THE IDEA Counts N from a radioactive source are recorded over time t. What should you plot to get a straight line?	ln N against t; gradient −k, intercept ln N_0. Compare with y = mx + c.
SPOT THE IDEA You suspect $y = a x^n$ but do not know n. What plot reveals n?	log y against log x; gradient n. This is a straight line with y-variable log y, x-variable log x and gradient n. Social scientists will know this as the log–log (elasticity) model.
SPOT THE IDEA A student solves $E = \frac{1}{2}mv^2$ for v and writes $v = \sqrt{(2E)} \div m$. What went wrong?	The m should be inside the square root. Dividing by m happens before the root is taken, so m ends up inside the root.
SPOT THE IDEA From 1/f = 1/u + 1/v, a student 'flips everything' and writes f = u + v. Why is this wrong?	The reciprocal of a sum is not the sum of the reciprocals. You may take the reciprocal of a whole side, never of each term separately. Test with numbers: 1/2 + 1/2 = 1, but 2 + 2 is not 1.
SPOT THE IDEA Solving $x^2 = 9$, a student writes x = 3. What is missing?	The solution x = −3. Squaring destroys sign information, so the square root brings back two candidates. Context decides which to keep: a length must be positive, a displacement need not be.
QUIZ In $T = 2\pi\sqrt{(L/g)}$, which operation do you undo first when solving for g?	The multiplication by 2π. 2π was applied last when building T from g, so it is the outermost layer and comes off first.
QUIZ From 1/f = 1/u + 1/v, which is the correct expression for v?	v = uf/(u − f). 1/v = 1/f − 1/u = (u − f)/uf; only then flip both sides.
QUIZ Make x the subject of ax + b = cx + d.	x = (d − b)/(a − c). Collect: ax − cx = d − b. Factor: x(a − c) = d − b. Divide.
QUIZ Solve $N = N_0e^{(-kt)}$ for t.	t = ln(N_0/N)/k. Divide by N_0, take ln, divide by −k and flip the fraction inside the log.
QUIZ Gravitational force goes as $1/r^2$. If r is halved, the force is multiplied by	4. $(\frac{1}{2})^{-2} = 4$.

QUIZ You measure decay counts N over time t. Which plot gives a straight line?	ln N against t. $\ln N = \ln N_0 - kt$: gradient $-k$, intercept $\ln N_0$.
QUIZ A student rearranges $E = \frac{1}{2}mv^2$ to get $v = \sqrt{(2E)/m}$. Which check exposes the error fastest?	Units: the right side isn't a speed. $\sqrt{(2E)/m}$ does not have units of m/s; at $E = 0$ both forms give 0, so that check misses it.
QUIZ Solving $x^2 = 16$ for a displacement that could be either direction, the full answer is	$x = \pm 4$. Squaring hides the sign, so both roots are candidates.